

BIO 4AE3 - 27 Oct 2025

[understanding the evolution (+ pop history) of Mendelian alleles involved in disease res/tolerance]

<u>qualitative</u> (Mendelian) coding sequence <u>RQ</u>	vs <u>quantitative traits</u> (quantitative genetics) [infinite alleles model] • regulatory/expression
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what maintains POLYMORPHISM?

• DEleterious REcessive •

hypotheses

- genetic drift (null)
 - size of pop?
 - HISTORIC size of pop? *
 - FOUNDER effects/bottlenecks
 - strength of selection / fitness cost
- heterozygote advantage //

TAY-SACHS disease

lethal abnormality → infant/toddler death

ALLELE freq -

~ 1/300 in US pop'n.

~ 1/30 in E European Jews

also. French Canadians, Cajuns.
Pennsylvania Dutch.

↑ pop gen suggests drift •

• RIDICULOUS speculation

PHENYLKETONURIA •

• phenylalanine metabolism

many different mutations

• PKU alleles are OLD •

• Populations without a history of being small •

Ireland ~ 2500 BC • pop size 100K-200K
current frequency is about 2x as expected •

HEterozyGOTE advantage?

need advantage of 1.5%

- mechanism/physiology?
 - higher phenylalanine
 - ~ higher birth weight?
 - ~ mycotoxin resistance?
 - ~ starvation resistance?